

BORUSAN MANNESMANN MATERIAL TEST REPORT

	HFW STEEL PIPE GENERAL INFORMATION																
ITEM	O.D. (in)	Wall Thickness (in)	Length (ft)	Specification or Reference and Grade	Coating	End Finish	NDT Weld Seam	NDT Pipe Body	Drift	Visual and Dimension Control	Flattenning Test	Min Collapse Pressure	Hydrostatic Test		Heat Treatment	Work Order	Date
													Pressure (Psi)	Hold Time (sec)			
1	8.625	0.322	38.0 - 47.0	API 5L 46th ed X52M PSL2	Zero VOC Clear Mill Coating	30° Beveled End	UT: OK N 10: 0.125in through hole	API 5L - E3.2 Pipe End Inspection - Welded Pipe	N/A	OK	OK	N/A	2970	5	Weld Seam Annealed Min 1598° F	W015018	11/17/2023

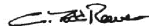
ITEM	Heat No	Lot No	Chemical Analysis														Ceq ^c		
				C (%)	Mn (%)	P (%)	S (%)	Si (%)	Al (%)	Cu (%)	Ni (%)	Cr (%)	Mo (%)	V (%)	Nb (%)	Ti (%)		B (%)	
			Max	0.220	1.400	0.025	0.015	0.450		0.200	0.200	0.100	0.150	0.050	0.050	0.040			0.250
			Min												Nb + V + Ti ≤ 0.15%				
			Ceq (sum)	C %	Mn % / 20			Si % / 30		Cu % / 20	Ni % / 60	Cr % / 20	Mo % / 15	V % / 10		5 X B%			
1	SDX-A313603	N/A	Product	0.068	0.630	0.012	0.000	0.238	0.018	0.131	0.047	0.076	0.015	0.000	0.024	0.013	0.000	0.120	
			Product	0.067	0.630	0.012	0.000	0.240	0.018	0.133	0.046	0.076	0.015	0.000	0.024	0.013	0.000	0.119	
			Heat	0.060	0.620	0.013	0.002	0.220	0.019	0.140	0.050	0.070	0.010	0.001	0.028	0.016	0.000	0.110	
1	SDX-A313605	N/A	Product	0.070	0.610	0.011	0.001	0.227	0.015	0.140	0.048	0.077	0.015	0.000	0.023	0.013	0.000	0.121	
			Product	0.069	0.610	0.011	0.001	0.227	0.015	0.141	0.048	0.077	0.015	0.000	0.023	0.012	0.000	0.120	
			Heat	0.060	0.600	0.011	0.002	0.210	0.016	0.150	0.050	0.070	0.010	0.002	0.027	0.015	0.000	0.110	
1	SDX-A313607	N/A	Product	0.073	0.620	0.012	0.001	0.224	0.014	0.138	0.048	0.082	0.015	0.000	0.023	0.013	0.000	0.124	
			Product	0.072	0.620	0.012	0.001	0.224	0.015	0.137	0.048	0.082	0.015	0.000	0.022	0.013	0.000	0.123	
			Heat	0.060	0.600	0.012	0.002	0.210	0.017	0.140	0.050	0.070	0.010	0.001	0.025	0.015	0.000	0.109	

ITEMHeat NoLot No			Mechanical Properties																
			Transverse Weld								Transverse Body								
			Seam Weld, 8" overall, 2.250" reduction area, 2" gauge length. Flat specimen		Impact Test at 32° F (ft/lb)					Pipe Body, 10.250" over all length, 2.250" length reduction area, 2" gauge length, T180, Flat specimen			Impact Test at 32° F (ft/lb)/ 3T90						
			Tensile Strength (PSI)	Rockwell Hardness (HRB)	Sample Size:	Full				Yield Strength (PSI)	Tensile Strength (PSI)	Elongation (%)	Rockwell Hardness (HRB)	Sample Size:	Full				
						Specimen 1 Impact Energy/ Shear	Specimen 2 Impact Energy/ Shear	Specimen 3 Impact Energy/ Shear	Average Impact Energy						Average Shear (%)	Specimen 1 Impact Energy/ Shear	Specimen 2 Impact Energy/ Shear	Specimen 3 Impact Energy/ Shear	Average Impact Energy
			Max	99								76,900			99				
Min	66,700	20	20	20					52,200	66,700	25		20	20	20				
1	SDX-A313603	N/A	Product	71,450	77	107.9 / 100	119.0 / 100	107.7 / 100	112	100	58,880	67,980	42	78	88.0 / 100	97.4 / 100	102.7 / 100	96	100
1	SDX-A313605	N/A	Product	70,510	74	81.4 / 100	94.3 / 100	105.3 / 100	94	100	55,090	67,940	43	71	102.3 / 100	78.4 / 100	82.8 / 90	88	97
1	SDX-A313607	N/A	Product	68,490	80	68.4 / 100	83.1 / 100	92.6 / 90	81	97	60,230	67,590	41	77	79.7 / 100	77.9 / 100	78.3 / 90	79	97

Coil Source: Steel Dynamics (1945 Airport Rd, Columbus, MS 39701)
Melted and Manufactured in USA

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WE CERTIFY THE ABOVE MATERIAL HAS BEEN MANUFACTURED, PROCESSED, SAMPLED, TESTED, AND INSPECTED TO MEET THE REQUIREMENTS OF THE STANDARD IN REFERENCE.
DETERMINATION OF MECHANICAL TESTING ARE ACCURATE AND TRACEABLE TO NIST AND CONFORM TO APPLICABLE TESTING METHODS.
CONFORMS TO NACE MR-0175 FOR HARDNESS ONLY AS IT APPLIES TO API PRODUCTS. MAX HARDNESS DOES NOT EXCEED RC22.
NOT VALID WITH INADEQUATE USE OF PIPE.
THIS INSPECTION CERTIFICATE IS IN ACCORDANCE WITH API 5L 46TH EDITION.

Todd Reeves/

VP OF QUALITY ASSURANCE